Week 13 Lecture
Review and Final Exam Information

IFN647

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Learning Objectives

Topic ID	Key topics	Mapping to slides
1	Final Exam Information	2-4
2	Review of IFN647 Teaching Strategies	5-8
3	Module 1 - Textual data pre-processing, IR, IE and Text Indexing	9-13
4	Module 2 - Language Models, Information Needs and Evaluation	14-18
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1. Final Exam Information

- IFN647 final exam is an on-campus invigilated exam in the end of semester exam period and deferred exam period.
- This semester's final exam will have 13 questions (a total of 45 marks)
  - **Short answer questions** (5 questions – 10 marks): Question 1 to Question 5
    - Answer question clearly in several rows in the in the booklets provided
    - Points are indicated
  - **Problem solving questions** (8 questions – 35 marks): Question 6 to Question 13
    - Write your answer clearly in the booklets provided.
    - A few questions need short python code.
    - Difficulty levels and question distribution (a normal distribution)
      - Peaceful or Easy
      - Normal or Neutral
      - Hard or Challenging

Exam Information cont.

- Need to know concepts, ideas and algorithms about textual data preprocessing, IR, IE, text indexing, language models, information needs and evaluation, learning to rank, user & relevance models, text classification & clustering, and Web search.
- Week 2 to week 12 review questions, and lecture notes.
- Please note that the exam paper tries to cover each week and avoid checking the same knowledge or concepts repeatedly.
- A few question requires writing python code. Don’t have to write much code usually and the basic python commands are enough.

Exam Questions

- Use weekly review questions (Week 2 to Week 12, excluding optional questions) and solutions to help you to prepare the final exam. Lecture notes are important references.
- The format of the exam questions is similar with those review questions, from which you can learn what concepts and knowledge you need to know.
- Know how to answer all review questions.
- Try it out by yourself and then compare your answers with the solutions.
- Contact the unit coordinator if you have any problems for preparing the final exam.
- Below is the lecture review, which includes some sample questions designed based on the review questions.

2. Review of IFN647 Teaching Strategies

- Lecturers choose **what** students will learn (based on curriculum design), and **how** the students will be assessed (e.g., assessment items and making guides).
  - This unit is more difficult than the others as it covers substantial technical content.
  - Describe complex technical details or ideas in mathematical equations and Python definitions.
  - Collect feedback from emails, workshops, lecture support sessions, IFN647 slack and tutors to support the teaching in this semester.
- How will students learn?
  - Read announcement, weekly overview, lecture notes
  - Attend online lecture or Watch Lecture Recordings
  - Attend workshop (on campus or online) and complete all tasks
  - Complete weekly review questions; and
  - Two assignments.
- To make you are confident for this unit, we used teacher-centred + learner centred pedagogy to "shift from teacher centred to learner centred pedagogy".
  - Have you developed strong, professional-level proficiency in applying ML techniques to NLP tasks?

What students will learn

- Review Syllabus planned by the unit coordinator.

Wk	Commencing	Lecture & Review Questions	Workshop
2	02/03/26	Text representation and preprocessing text	Pre-processing Textual Data
3	09/03/26	Information retrieval, word embedding & generative models	Pre-Processing: Stemming and Python Classes
4	16/03/26	Index text documents and information extraction	IR models and tf*idf
5	23/03/26	Language models & evaluation metrics	Indexing and Query Processing
6	30/03/26	User information needs and ranking effectiveness	Language models
7	13/04/26	Learning to Rank	Evaluation
8	20/04/26	Relevance Models & Social Media Analytics	Information Needs and Training Data Generation
9	27/04/26	Text Classification, Feature Selection and Sentiment Analysis	Learning to Rank
10	04/05/26	No Lecture	Relevance (Pseudo) Models vs. IR models
11	11/05/26	Text Clustering and Recommender Systems	Text classification & clustering
12	18/05/26	Web Search, Feature Dependencies and Multimodal Learning	Lab drop-in

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What have you learned?

- You may develop your own understanding of the material you have learned, for example:

Traditional NLP Tasks	Text processing & parsing, IR and Web search, Text Indexing, and Information Extraction
Machine Learning for NLP	Language models, Learning to rank, Semi-supervised learning, Text classification & Clustering
Real-World NLP Challenges	Understanding user information needs, Improving ranking effectiveness, Sentiment analysis, Topic modeling (e.g., LDA), Recommender systems
Advanced NLP Techniques	Pseudo relevance techniques (KL divergence), Contextual embeddings (e.g., BERT), Dual-Encoders, LDA-based clustering, Attention mechanisms for modeling feature dependencies, Multimodal learning, Contrastive learning
NLP in the Era of LLMs	Integration of pre-trained foundation models into NLP pipelines (e.g., “bert-base-uncased”), Few-shot and zero-shot learning capabilities of LLMs, and Enhanced reasoning (CoT, ABSA, MAST) & generation (RAG), and interaction abilities (e.g., Two-stage LLM driven clustering, RecPrompt).

Australia Tech and IT Job Market Is Changing?

- **Key Trends -** Entry-level barriers are increasing in the IT job market
  - Greater emphasis on automation, with AI replacing many junior-level tasks (e.g., testing and routine work)
  - Strong demand in high-growth areas: Cybersecurity, Cloud infrastructure, Data & AI, and Platform engineering
- **How to Stay Competitive as a Candidate**
  - Keep your professional profile and skills up to date
  - Maintain regular communication with recruiters
  - Build visibility through networking and industry engagement
- **Upskilling & Career Development**
  - Demand remains strong for specialised (niche) talent
  - Develop deep expertise in high-demand domains
    - Consider advanced qualifications:
      - PhD / MPhil
    - Engage in industry–academia collaborations (e.g., projects with QUT)

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3. Module 1 – Text processing & indexing, IR and IE

- From Week 2 – Week 4

Key topics in week 2
Text Representation
Precocessing text: *Text Statistics & Document Parsing*
Noise Reduction: *Stopping and Stemming*
Feature Extraction: *Phrases & N-grams*
Document Structure and Markup: HTML tags & Hyperlinks

Key topics in week 3
Overview of IR Models
Vector Space Model, e.g., tf*idf
Word embedding
Generative Models: Bayes classifier
Binary Independence Model
The BM25 Ranking Algorithm
Abstract Model of Ranking for IR and ML: $R(Q, D) = g_1(Q)f_1(D) + g_2(Q)f_2(D) + \ldots + g_n(Q)f_n(D)$

Key topics oin week 4
Indexes and text representation
Index Construction
Query Processing: Document-at-a-time and Term-at-a-time
Information extraction
Hidden Markov Model
LLM-Based Information Extraction

Sample Question from week 2

Sample **<u>Question 1</u>** (Short Answer)

Which of the following is FALSE? and explain why it is FALSE. # wk2, Q2

(a) Stemming is a component of text processing that captures the relationships between different variations of a word.

(b) Stemming reduces the different forms of a word that occur because of inflection (e.g., plurals, tenses) or derivation (e.g., making a verb into a noun by adding the suffixation) to a common stem.

(c) In general, using a stemmer for search applications with English text produces a small but noticeable improvement in the quality of results.

(d) A dictionary-based stemmer uses a small program to decide whether two words are related, usually based on knowledge of word suffixes for a particular language.

Question 1 Answer: (*<u>Understand the difference between two concepts</u>*)

(d)

A dictionary-based stemmer *has no logic of its own, but instead relies on pre-created dictionaries of related terms to store term relationships. In contrast,* ***an algorithmic stemmer*** *uses a small program to decide whether two words are related, usually based on knowledge of word suffixes for a particular language.*

Sample Question from week 3

Sample **Question 2** (Bayes Classifier) # similar with wk3, Q2

[Total marks for Question 2 = 4]

Let Q = {US, ECONOM, ESPIONAG} be a query, and $C = \{D_1, D_2, D_3, D_4, D_5, D_6, D_7\}$ be a collection of documents. The following tables shows each document's terms and relevance to Q:

Document ID	Terms: d_{ij}	Relevance to Q
D_1	GERMAN, VW	0 no
D_2	US, US, ECONOM, SPY	1 yes
D_3	US, BILL, ECONOM, ESPIONAG	1 yes
D_4	US, ECONOM, ESPIONAG, BILL	1 yes
D_5	GERMAN, MAN, VW, ESPIONAG	0 no
D_6	GERMAN, GERMAN, MAN, VW, SPY	0 no
D_7	US, MAN, VW	0 no

Assume document D = {US, ECONOM, ESPIONAG}, terms are independent and the estimate of $P(D|R)$ can be calculated using the following question:

$$P(D|R) = \prod_{i=1}^{t} P(d_i|R)$$

(a) Let T = {US, BILL, ECONOM, ESPIONAG, GERMAN, MAN, VW, SPY} = $\{d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8\}$. Then D can be represented as {US, ECONOM, ESPIONAG} = $\{d_1, d_3, d_4\}$, and its binary vecytor is (1, 0, 1, 1, 0, 0, 0, 0). Write the binary vectors of the three relevant documents.

2 marks

(a) Calculate $P(D|R)$.

2 marks

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Sample Question 2 solution

Question 2 Solution:

(a)

The three relevant documents are D_2, D_3 and D_4 with the following binary vectors:

$D_2 = (1, 0, 1, 0, 0, 0, 0, 1)$
$D_3 = (1, 1, 1, 1, 0, 0, 0, 0)$
$D_4 = (1, 1, 1, 1, 0, 0, 0, 0)$

(b)

Let $p_i = P(d_i|R)$ be the probability that term d_i occurs (i.e., has value 1) in relevant documents, so we have

$p_1 = P(d_1|R) = P(\text{'}US\text{'}|R) = 3/3 = 1.0$

$p_3 = P(d_3|R) = P(\text{'ECONOM'}|R) = 3/3 = 1.0$

$p_4 = P(d_4|R) = P(\text{'ESPIONAG'}|R) = 2/3 = 0.667$

By using the following equation, we have

$P(D|R) = P(d_1|R) * P(d_3|R) * P(d_4|R) = 1 * 1 * 0.667 = 0.667$

Sample Question from week 4

Sample **QUESTION 3** (short answer) #wk4, Q1

Which of the following is wrong? justify your answer.

(a) The index is inverted because usually we think of words being a part of documents, but if we invert this idea, documents are associated with words.

(b) In an inverted index that contains only document information, i.e., the features are binary, meaning they are 1 if the document contains a term, 0 otherwise. This inverted index contains enough information to tell if the document contains the exact phrase "tropical fish".

(c) Each index term has its own inverted list that holds the relevant data for that term. Each list entry is called a posting, and the part of the posting that refers to a specific document or location is often called a pointer.

(d) An extent is a contiguous region of a document. We can represent these extents using word positions.

Question 3 Answer: (*Understand topical features' representation*)

(b)

Although a document that contains the words "tropical" and "fish" is likely to be relevant; however, we really want to know if the document contains the exact phrase "tropical fish". To do this, we need to add position information to the index.

Module 2: Language Models, Information Needs and Evaluation

- Week 5 & Week 6

Key topics in week 5
Introduction of Language Models
Query-Likelihood Model
Smoothing Technique: e.g., JM smoothing
Evaluation Overview and Relevance Judgments
Effectiveness Measures: e.g., precision, recall, F1
Efficiency Metrics
Significance Tests

Key topics in week 6
Information filtering and Information Needs
Query-Based Methods for Capturing Information Needs: e.g., Noisy Channel Model
Relevance feedback using machine learning
Ranking Effectiveness: e.g., MAP, DCG & NDCG
Interpolation
Discounted Cumulative Gain

Sample Question from week 5

Sample **<u>QUESTION 4</u>** (Language Models) # wk5, Q1

[Total marks for Question 4 = 4]

Let Q = {US, ECONOM, ESPIONAG} be a query, and $\boldsymbol{C} = \{D_1, D_2, D_3, D_4, D_5, D_6\}$ be a collection of documents, where

D_1 = {GERMAN, VW}

D_2 = {US, US, ECONOM, SPY}

D_3 = {US, BILL, ECONOM, ESPIONAG}

D_4 = {US, ECONOM, ESPIONAG, BILL}

D_5 = {GERMAN, MAN, VW, ESPIONAG}

D_6 = {GERMAN, GERMAN, MAN, VW, SPY}

Jelinek-Mercer smoothing method uses the following equation to calculate a score for a document D

$$P(Q|D) = \prod_{i=1}^{n} p(q_i|D)$$

where n is the number of query terms in Q, and

$$p(q_i|D) = (1-\lambda)\frac{f_{q_i,D}}{|D|} + \lambda\frac{c_{q_i}}{|C|}$$

for all query term q_i in Q, where $f_{qi,D}$ is the number of times query word q_i occurs in document D, $|D|$ is the number of word occurrences in D, c_{qi} is the number of times query word q_i occurs in collection C, and $|C|$ is the total number of word occurrences in collection C. Assume parameter $\lambda = 0.4$, calculate $P(Q|D_2)$. You are required to write the finding process.

Sample Question 4 solution

(Understand the meaning of variables in equations and use equations in practical examples)

Question 4 Solution:

D_2 = {US, US, ECONOM, SPY}, Q = {US, ECONOM, ESPIONAG}

P(Q| D_2) = P(US | D_2) * P(ECONOM | D_2) * P(ESPIONAG | D_2) = 0.003865

P(US | D_2)=(1-0.4)*(2/4)+0.4*(4/23)=0.6*0.5 + 0.4*0.17 = 0.368

P(ECONOM | D_2) =0.6 *(1/4)+0.4*(3/23) = 0.6*0.25 + 0.4*0.13 = 0.202

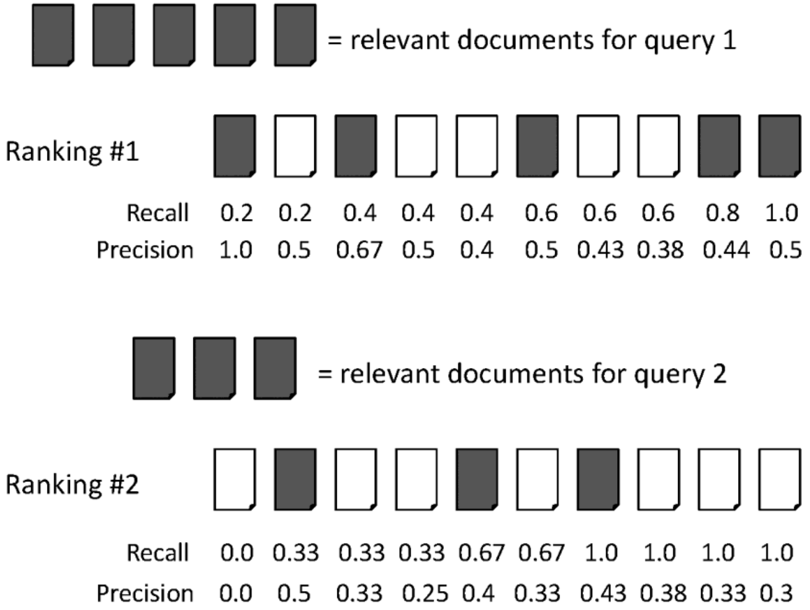
P(ESPIONAG | D_2) = 0.6*(0/4) + 0.4*(3/23) = 0.4*0.13 = 0.052

Sample Question from week 6

Sample **QUESTION 5** (Evaluation) # Similar with wk6, Q4

[Total marks for Question 5 = 3]

MAP (Mean Average Precision) summarizes rankings from multiple queries by averaging average precision. The following are two ranking results for two queries. Calculate the MAP for these rankings. You are required to write the calculating process.



Sample Question 5 solution

Solution for QUESTION 5:

$$average\ precision\ query\ 1 = (1.0 + 0.67 + 0.5 + 0.44 + 0.5)/5 = 0.62$$
$$average\ precision\ query\ 2 = (0.5 + 0.4 + 0.43)/3 = 0.44$$

$$mean\ average\ precision = (0.62 + 0.44)/2 = 0.53$$

5. Module 3 – Learning to rank, User & Relevance Models

- Week 7 & week 8

Key topics in week 7
Learning to Rank
Query expansion
Semi-supervised learning for Learning to Rank
Rocchio-Based Information Filtering Model
Probabilistic Information Filtering Models
BM25-based IF model
LLM-based Information Filtering

Key topics in week 8
User Models
Relevance Models
Pseudo Relevance Models
Social Media Analytics
Social Search
Traditional IR Models vs. LLMs in Social Search

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Sample Question from week 7

Sample **QUESTION 6** (**Mutual Information**) #similar with wk7 Q1

[Total marks for Question 6 = 4]

For two words (or terms) *a* and *b*, their mutual information measure (*MIM*) is defined as

$$\log \frac{P(a, b)}{P(a)P(b)}$$

where *P*(*a*) is the probability that word *a* occurs in a text window of a given size, *P*(*b*) is the probability that word *b* occurs in a text window, and *P*(*a*, *b*) is the probability that *a* and *b* occur in the same text window.

Assume a function *df*(*docs*, *a*) is defined to calculate the number of documents containing word *a*, and function *df*2(*docs*, *a*, *b*) returns the number of documents containing both words *a* and *b*, where *docs* is a set of documents.

Let *D* be a set of documents. For any two words *a* and *b*, use the *MIM* equation and the provided two functions to calculate their *MIM*. You need to write math formulas (or equivalent python definitions) to compute *P*(*a*), *P*(*b*) and *P*(*a*, *b*) in terms of *D*, *df* and *df*2.

Sample Question 6 solution

Compute based on equations and function definitions

Question 6 Solution

Let N= |D|, then we have

$P(a)= df(D, a)/ N, P(b)= df(D, b)/N,$

$P(a, b) = df2(D, a, b)/N$

So, MIM = $log(df2(D, a, b)/N)/((df(D, a)/ N)*(df(D, b)/N))$

= $log[|D|*df2(D, a, b)/(df(D, a)*df(D, b))]$

Sample Question from week 7 cont.

Sample **Question 7** (Short Answer)

Which of the following descriptions is wrong? and justify your answer.

(a) Large language models (LLMs) can naturally generate human-readable explanations for recommendations, thereby improving system transparency and interpretability.

(b) LLM-based information filtering does not require labelled training data, as well-designed prompts can effectively guide the model's behaviour.

(c) Retrieval-Augmented Generation (RAG) is an AI framework that enhances LLMs by retrieving relevant information from external knowledge bases, leading to more accurate and factually grounded outputs.

(d) LLMs such as GPT-4 have demonstrated strong capabilities in natural language understanding, and their effectiveness in real-world news recommendation systems has been fully established.

(e) Chain-of-Thought (CoT) prompting is a method to improve LLMs by prompting them to generate step-by-step reasoning before giving a final answer.

Solution: (d)

LLMs such as GPT-4 have demonstrated strong capabilities in natural language understanding; however, their effectiveness in real-world news recommendation systems has yet to be fully established.

Sample Question from week 8

Sample **Question 8** (Short Answer)

Which of the following is FALSE? and explain why it is FALSE. # wk8, Q1

(a) In the query likelihood retrieval model, documents are ranked by the probability that the query text could be generated by the document language model.

(b) To use the unigram language model to estimate $P(Q|D)$ - the document D's score for the given query $Q = \{q_1, q_2, \ldots, q_n\}$, the language model probabilities $P(q_i|D)$ ($i = 1, 2, \ldots, n$) are needed to be estimated.

(c) The major problem with the estimate of $P(q_i|D)$ is that if any of the query words are missing from the document, the score given by the query likelihood model for $P(Q|D)$ will be zero.

(d) Kullback-Leibler (KL) divergence is a technique for avoiding this estimation problem and overcoming data sparsity, which means that we typically do not have large amounts of text to use for the language model probability estimates.

Question 8 Answer: (*Understand the concept of KL divergence*)

(d) *Kullback-Leibler (KL) divergence is a measure to estimate the difference between two probability distributions. Smoothing is a technique for avoiding this estimation problem and overcoming data sparsity, which means that we typically do not have large amounts of text to use for the language model probability estimates.*

Sample Question from week 8 cont.

Sample **QUESTION 9** (likelihood method) # wk8, Q4

[Total marks for Question 8 = 3]

Assume C is a collection of tweets. We can use the following query likelihood method to calculate the probability of a tweet d occurring for a given query Q.

$$P(Q,d) = \sum_{w \in Q,d} c(w,Q) \log\left(1 + \frac{c(w,d)}{\mu \times P(w|C)}\right) + |Q| \log \frac{\mu}{\mu + |d|}$$

Let $|d|$ and $|Q|$ are the respective lengths (size) of the tweet d and query Q, and μ is the smoothing parameter.

Interpret the meaning of $c(w, d)$, $c(w, Q)$, and $P(w|C)$.

Sample Question 9 solution

Understand definitions in a math equation

Question 9 solution

- $c(w, d)$ is the word w's counts in the given tweet d,
- $c(w, Q)$ is the word w's counts in the given query Q,
- $P(w|C)$ is the probability of the word w in the collection C that is used to normalise the model.

5. Module 4: Text classification and Clustering

Week 9 and week 11

Key topics in week 9
Text Classification
Feature Selection & Evaluation
Vector Space Classifiers
Generative Model - Naïve Bayes Classifier
Sentiment Analysis
Methodologies for Sentiment Analysis
Sentiment Analysis in the Era of LLMs

Key topics in week 11
Cluster Analysis – Unsupervised Learning
Clustering Methods
LDA (Model)-Based Clustering and Evaluation
LLM-Era Clustering
Recommender Systems
LLMs for News Recommendation

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Sample Question from week 9

Sample **QUESTION 10** **(Classification)** # similar with wk9, Q3

[Total marks for Question 9 = 6]

Let *C* be a set of classes and *D* be a training set, where each element in *D* is a pair (d_j, c_j), d_j is a document and c_j is its class. The training procedure of **Rocchio classification** for the input training set *D* uses *centroids* to represent classes. The centroid of a class is computed as the average vector of class members.

For the given topic "R101", we have obtained the following training set which includes seven documents, ten selected features (terms) (VM, US, Spy, Sale, Man, GM, Espionag, Econom, Chief, and Bill) and the corresponding term weights, where *Class* = 1 means relevant, and *Class* = -1 means irrelevant.

Doc	VM	US	Spy	Sale	Man	GM	Espionag	Econom	Chief	Bill	Class
'39496'	0.17	0.01	0.20	0.00	0.02	0.20	0.12	0.11	0.00	0.00	1
'46547'	0.10	0.21	0.10	0.00	0.00	0.00	0.22	0.20	0.00	0.10	1
'46974'	0.00	0.23	0.10	0.20	0.05	0.10	0.10	0.10	0.01	0.00	1
'62325'	0.17	0.01	0.20	0.00	0.02	0.20	0.12	0.11	0.00	0.00	1
'6146'	0.10	0.00	0.00	0.30	0.10	0.20	0.00	0.12	0.10	0.00	-1
'18586'	0.00	0.30	0.00	0.30	0.20	0.00	0.00	0.20	0.15	0.20	-1
'22170'	0.20	0.00	0.00	0.15	0.20	0.25	0.00	0.00	0.00	0.10	-1

Question 9 continued overleaf

Question 9 continued

Assume the above training set is represented as two Python lists as follows, where *X_train* is the list of document vectors of documents in *D*, and *y_train* is the list of the corresponding classes.

X_train =[[0.17, 0.01, 0.20, 0.00, 0.02, 0.20, 0.12, 0.11, 0.00, 0.00],
[0.10, 0.21, 0.10, 0.00, 0.00, 0.00, 0.22, 0.20, 0.00, 0.10],
[0.00, 0.23, 0.10, 0.20, 0.05, 0.10, 0.10, 0.10, 0.01, 0.00],
[0.17, 0.01, 0.20, 0.00, 0.02, 0.20, 0.12, 0.11, 0.00, 0.00],
[0.10, 0.00, 0.00, 0.30, 0.10, 0.20, 0.00, 0.12, 0.10, 0.00],
[0.00, 0.30, 0.00, 0.30, 0.20, 0.00, 0.00, 0.20, 0.15, 0.20],
[0.20, 0.00, 0.00, 0.15, 0.20, 0.25, 0.00, 0.00, 0.00, 0.10]]

y_train = [1, 1, 1, 1, -1, -1, -1]

Design a Python function *train_Rocchio*(*X_train*, *y_train*) to calculate the centroid of the relevant class (named as 'R101') and the centroid of irrelevant class (named as 'iR101'), respectively. For example, use the above table, we have 'iR101' = [0.100, 0.100, 0.000, 0.250, 0.167, 0.150, 0.000, 0.107, 0.083, 0.100].

Sample Question 10 solution

Question 10 Solution

```
import sys, os
def train_Rocchio( X, y):                                      (1 mark)
    pos_mean=[0,0,0,0,0,0,0,0,0,0]
    neg_mean = [0,0,0,0,0,0,0,0,0,0]
    pos_n = len([x for x in y if x==1])
    neg_n = len([x for x in y if x==-1])                       (1 mark)
    for i in range(len(y)):                                    (2 marks)
      if y[i] == 1:
        pos_mean = [pos_mean[j]+X[i][j] for j in range(10)]
      elif y[i] == -1:
        neg_mean = [neg_mean[j]+X[i][j] for j in range(10)]
    pos_mean = [pos_mean[j]/pos_n for j in range(10)]          (2 marks)
    neg_mean = [neg_mean[j]/neg_n for j in range(10)]
    return { 'R101': pos_mean, 'iR101': neg_mean }
```


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Sample Question from week 11

Sample **QUESTION 11** **(Distance function)** # similar with wk11, Q1

[Total marks for Question 10 = 4]

Table 1 shows an example of a binary information table.

(a) Draw the binary contingency table of Table 1 for comparing documents d_1 and $d_{2.}$

(b) Evaluate the Jaccard distance between d_1 and d_2.

(c) Evaluate the Simple matching coefficient distance between d_1 and d_2.

Document	t_1	t_2	t_3	t_4	t_5	t_6	t_7
d_1	1	1	0	0	0	0	0
d_2	0	0	1	1	0	1	0
d_3	0	0	1	1	1	1	0
d_4	0	0	1	1	1	1	0
d_5	1	1	0	0	0	1	1
d_6	1	1	0	0	0	1	1

Table 1. A binary information table

Sample Question 11 solution

- **Question 11 Solution**

(a) 2 marks

		Document d_2		
		1	0	Sum
Document d_1	1	0	2	2
	0	3	2	5
	Sum	3	4	7

(b) 1 mark

$disJac(d1, d2) = (r+s) / (q+r+s) = 5/5 = 1$

(c) 1 mark

$disS(d1, d2) = (r+s) / (q+r+s+t) = 5/7$

7. Module 5: Web Search, Feature Dependencies and Multimodal Learning

- Week 12

Key topics in week 12
Web search
Link analysis
Attention Mechanism for Feature Dependency
Multi-View Data
Multimodal Data
Contrastive Learning

Sample Question from week 12

Sample **Question 12 (PageRank)** #wk12, Q1

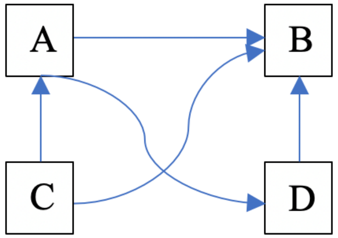
[Total marks for Question 11 = 4]

A Web graph $G = (P, L)$ consists of Web pages (vertices) and links (edges). The PageRank procedure takes a Web graph G as input and then outputs the better PageRank estimate PR using the following equation

$$PR(u) = \frac{\lambda}{N} + (1-\lambda). \sum_{v \in B_u} \frac{PR(v)}{L_v}$$

where B_u is the set of pages that point to u, and L_v is the number of outgoing links from page v.

The following figure $G = (P, L)$ is a Web graph, where P = {A, B, C, D} and L = {(A, B), (A, D), (C, A), (D, B), (C, B)}.



Assume the initial estimate of each Web page is equally, i.e., $PR(A) = PR(B) = PR(C) = PR(D) = 0.25$; and $\lambda = 0.15$. Calculate the PageRank estimate (PR value) of each Web page after running the first iteration of procedure PageRank(G).

Sample Question 12 solution

(Learn how to use equations in iterative calculations)

Question 12 Solution:

The first iteration:

$\mathbf{L}_C$=|{A, B}|=2, $\mathbf{L}_A$=|{B, D}|=2, $\mathbf{L}_D$=|{B}|=1, $\mathbf{L}_B$=|{}|=0,

For page A: $\mathbf{B}_A$ = {C}

PR(A) = 0.15/4 + (1-0.15)*(PR(C)/L_c) =0.0375 + 0.85*(0.25/2) = 0.14375

For page B: $\mathbf{B}_B$ = {A,C,D}

PR(B) = 0.15/4 + (1-0.15)*(PR(C)/2 + **PR(A)**/2 + PR(D)/1) = 0.0375 + 0.85*(0.25/2 + **0.25**/2 + 0.25) = 0.4625

For page C: $\mathbf{B}_c$ = {} # special case

PR(C) = 0.15/4 + (1-0.15)*0 = 0.0375

For page D: $\mathbf{B}_D$ = {A}

PR(D) = 0.15/4 + (1-0.15)*(PR(A)/2) = 0.0375 + 0.85*(0.25/2) = 0.14375

References

[1] IFN647 Canvas - week 2 to week 12 review questions

[2] IFN647 Canvas - week 2 to week 12 question solutions

[3] IFN647 Canvas - week 1 to week 12 lecture notes

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